

Alex Rivera

Machine Learning Engineer | alex.rivera@example.com | +1 555 010 2233 | github.com/alex-rivera-demo | linkedin.com/in/alex-rivera-demo

SUMMARY

Machine learning engineer with hands-on experience building and deploying predictive models and search systems. Passionate about turning data into products. Expert in machine learning and deep learning.

EDUCATION

B.Tech in Computer Science and Engineering - Northfield Institute of Technology

2021 - 2025 | CGPA: 8.7/10

Relevant coursework: Machine Learning, Data Structures and Algorithms, Database Systems, Distributed Systems, Information Retrieval.

EXPERIENCE

Machine Learning Intern - Finlytics

Jan 2025 - Jun 2025 | Fintech startup, Bengaluru

- Built a customer churn prediction system using XGBoost that achieved an AUC of 0.91 on the holdout set.
- Engineered 40+ behavioural features from 14 months of transaction and support-ticket data for 120,000 customers.
- Deployed the model as a weekly batch-scoring pipeline with Airflow and a FastAPI endpoint used by the retention team.
- Helped reduce customer churn by 18% through targeted retention campaigns.
- Used SHAP to explain model predictions to non-technical stakeholders.

Software Engineering Intern - CartWave

May 2024 - Jul 2024 | E-commerce platform

- Developed REST APIs in Python (FastAPI) for the order-tracking service handling 2M+ requests per day.
- Introduced Redis caching for product lookups, reducing p95 latency by 40%.
- Worked on various backend improvements and bug fixes.
- Wrote unit and integration tests with pytest, raising test coverage from 52% to 81%.

PROJECTS

LexiSearch - Semantic Search for Legal Documents

Python, sentence-transformers, FAISS, FastAPI, React

- Built a semantic search engine over 25,000 court judgments using sentence embeddings and a FAISS index.
- Achieved MRR@10 of 0.72 versus 0.51 for a BM25 baseline, with p95 query latency of 120 ms.
- Implemented chunking, metadata filters and a React UI with highlighted passages.

PulseCheck - Real-time Anomaly Detection for IoT Sensors

PyTorch, Kafka, Docker

- Trained an LSTM autoencoder to flag anomalies in streaming temperature and vibration data.
- Processed 5,000 events per second through Kafka with sub-second alerting.
- Designed a scalable, fault-tolerant architecture.

Campus Marketplace - Full-stack Web App

Next.js, TypeScript, PostgreSQL, Prisma

- Led a team of 4 to build a buy/sell marketplace used by 1,200+ students.
- Implemented authentication, listing search and an admin moderation dashboard.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java

ML / AI: PyTorch, scikit-learn, XGBoost, pandas, NumPy, SHAP, sentence-transformers, FAISS, Hugging Face Transformers

Backend and data: FastAPI, Node.js, PostgreSQL, Redis, Kafka, Airflow

Cloud and DevOps: Docker, AWS (EC2, S3), GitHub Actions, familiar with Kubernetes

ACHIEVEMENTS

- Winner, HackNorth 2024 (Best Use of AI) - built an accessibility assistant for visually impaired users in 36 hours.
- Co-author, "Hybrid Retrieval for Low-Resource Legal Search", Student Research Symposium 2024.
- Top 5% in the Kaggle "Tabular Playground" competition.